

Lab 10: Modeling for Decision Making

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Objective: In this lab, students will try to propose a solution to an open-ended problem based on their respective understanding of the given scenario.

Instructions: Below are some instructions. Please go through them carefully:

- Given below is a scenario where students need to propose a solution-based
- Use the traditional approach to name your files for submission:
 - `ROLLNUMBER_IE507_Lab10.ipynb`;
 - `ROLLNUMBER_IE507_Lab10_Report.pdf`

Question 1: Efficient Tour Planning- A Heuristic Approach

Consider that **S1** and **S2** are two cities miles apart as hometown and workplace (are known as source nodes, **S**). A list of cities of attractions for your itinerary is given as a set **C** (representing the cities you plan to visit along with your parents). Further, these cities of attractions are distributed over a geographical region, and direct connectivity may/may not be possible from one tourist attraction to the other. Both **S1** and **S2** belong to metropolitan regions, and a direct route is possible from **S** to **C**. But you are constrained by choice to select a meeting point from a set **M**, which intersects on **S1 – S2** route. All cities in **C** are well spread and may be connected from **M** via. some distance and trains respectively. Below is a graph in Fig. 1 for the network connecting various cities in **M** and **C**. Note that a dashed-blue vertex is between the edges of elements of **C** with a meeting point in **M**. A black vertex represents the route link between a meeting point in set **M** and a source city from **S**. Below is a sample layout for a graph with two source nodes, one meeting points, and one city of attractions. Further, we provide you with some datasets to describe the characteristics of the below network.

Data Set ($\forall i, j \in S \cup C \cup M$)

- **S** = {**S1**, **S2**} represents source nodes
- **C** = {**C1**, **C2**, ..., **Cp**} represents nodes for cities of attraction where $p = 1$
- **M** = {**M1**, **M2**, ..., **Mq**} represents nodes for possible meeting points where $q = 1$
- **N** = $[n]_{ij}$ denotes the *adjacency matrix* representing the possible route between node i to node j .

	S1	S2	M1	C1
S1	0	0	1	0
S2	0	0	1	0
M1	1	1	0	1
C1	0	0	1	0

- **D** = $[d]_{ij}$ denotes the *distance matrix* representing the distance (in kms) between node i to node j .

	S1	S2	M1	C1
S1	0	0	1508	0
S2	0	0	807	0
M1	1508	807	0	167
C1	0	0	167	0

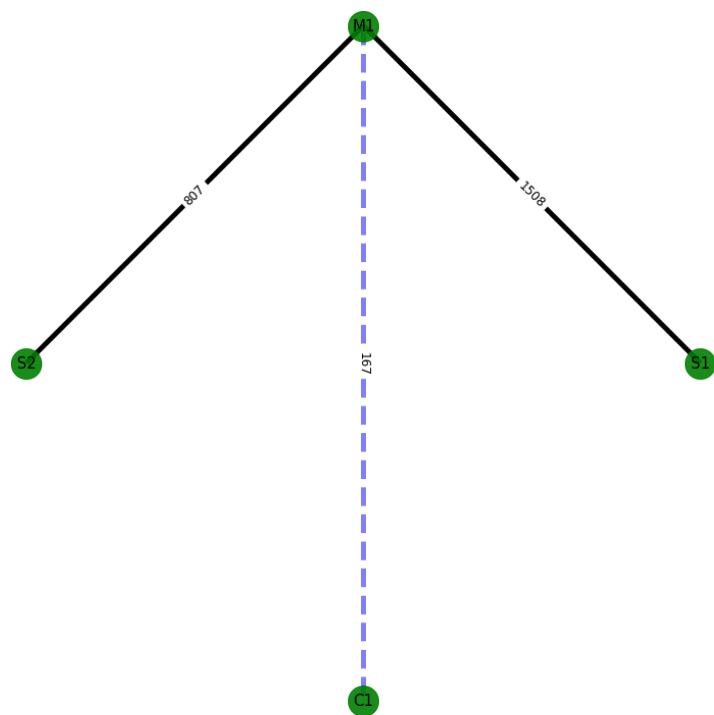


Figure 1: A layout for network with two source cities (**S1**,**S2**), one meeting point **M1** and one city of attraction **C1**

- $\mathbf{F} = [\mathbf{f}]_{ij}$ denotes the *frequency matrix* representing the daily frequency of trains between node i to

		S1	S2	M1	C1
node j .	S1	0	0	8	0
	S2	0	0	5	0
	M1	8	5	0	6
	C1	0	0	6	0

- $\mathbf{TT} = [\mathbf{tt}_k]_{ij}$ denotes the *travel time matrix* representing the travel time for k^{th} train between node i to node j where $\mathbf{k} \in \mathbf{F}_{ij}$. Note that travel time is symmetric that is travel time for node j to node i is same as node i to node j .

S1 to M1: [18:35, 21:27, 19:46, 26:25, 28:36, 20:28, 32:16, 22:10]

S2 to M1: [16:25, 19:38, 20:56, 09:13, 12:08]

M1 to C1: [03:42, 02:56, 04:15, 03:18, 05:08, 03:50]

- $\mathbf{DEP} = [\mathbf{dep}_k]_{ij}$ denotes the *departure time matrix* representing the departure time for k^{th} train between node i to node j where $\mathbf{k} \in \mathbf{F}_{ij}$.

S1 to M1: [16:30, 14:40, 03:55, 01:25, 10:26, 12:25, 06:35, 20:15]

M1 to S1: [18:25, 06:20, 02:25, 05:15, 09:20, 15:15, 20:18, 22:20]

S2 to M1: [04:20, 08:35, 12:15, 18:12, 22:18]

M1 to S2: [12:25, 15:16, 18:21, 07:25, 21:20]

M1 to C1: [05:50, 10:25, 12:30, 18:15, 21:05, 16:20]

C1 to M1: [04:00, 07:15, 10:55, 18:35, 21:25, 23:50]

- $\mathbf{ARR} = [\mathbf{arr}_k]_{ij}$ denotes the *arrival time matrix* representing the arrival time for k^{th} train between node i to node j where $\mathbf{k} \in \mathbf{F}_{ij}$.

NOTE: Model the scenario below using your programming skills so that your proposed solution works for large networks. For simplicity, assume that there is zero stoppage time at each node. Also, all the train timetable is in a 24-hour format.

Scenario C: Parting Ways

Assuming both passengers board the same train from **C1** to return to their destinations for $\{\mathbf{S1}, \mathbf{S2}\}$, but at **M1** they part their respective ways. A layover for both passengers can be expected on return as well, followed by a waiting time for passenger departing later from **M1**. In this scenario, a layover can be computed as the time between the arrival of a train at **M1** from **C1** and the time of the first passenger departure. Following layover time, a waiting time for other passengers is the time he waits for his train. Consider a minimum layover time of 1 hour and a maximum waiting time of 2 hours.

1. Based on the given feasibility conditions, plot a multiple bar graph for feasible Out-bound trains from **M1** to $\{\mathbf{S1}, \mathbf{S2}\}$ against Inbound trains from **C1** to **M1**. (You can use different colours for S1 and S2, look at the hint with a sample code for more clarity).
2. Suppose we want to minimise the total travel time of both passengers. The aim is to minimize $2 \times (\text{travel time from C1 to M1}) + (\text{travel time from M1 to S1}) + (\text{travel time from M1 to S2}) + (\text{waiting time}) + (\text{layover time})$. Do an exhaustive search to print the top 3 feasible travel plans (3 travel plans taking minimum total time). Remember, you do not have to model this as an optimization problem, but you

have to do a search. (output should look like **From C1 to M1 take train X; from M1 to S1 take train Y; from M1 to S2 take train Z; total travel time = AA hours BB mins**).

- repeat the above two parts when minimum layover time is in set $\{1,2\}$ hours and the maximum waiting time is in set $\{2,5,10\}$ hours.

Hint: Plotting a multiple bar graph using matplotlib

```
import numpy as np
import matplotlib.pyplot as plt

X = ['Train A', 'Train B', 'Train C', 'Train D']
m1s1= [10,20,20,40]
m1s2 = [20,30,25,30]

X_axis = np.arange(len(X))
width=0.3

plt.bar(X_axis - width/2, m1s1, width=width, label = '# feasible trains M1 to S1')
plt.bar(X_axis + width/2, m1s2, width=width, label = '# feasible train M1 to S2')

plt.xticks(X_axis, X)
plt.xlabel("trains from C1 to M1")
plt.ylabel("Number of trains")
plt.legend()
plt.show()
```

Scenario D: Introducing Direct Trains from Source to Cities to City of Attractions

Railways has introduced two direct trains from **S1** to **C1** and **S2** to **C1** and return trains as well, with details given below:

- S1** → **M1** → **C1**(05:30-S1C1.05:15): Departing **S1** @ 05:30, reaching **M1**@01 : 30 (+1 day) and finally arriving **C1**@05 : 15 (+1 day).
- S1** → **M1** → **C1**(20:15-S1C1.22:15): Departing **S1** @ 20:15, reaching **M1**@18 : 15 (+1 day) and finally arriving **C1**@22 : 15 (+1 day).
- S2** → **M1** → **C1**(15:45-S2C1.08:45): Departing **S2** @ 15:45, reaching **M1**@05 : 45 (+1 day) and finally arriving **C1**@08 : 45 (+1 day).
- S2** → **M1** → **C1**(07:30-S2C1.00:45): Departing **S1** @ 07:30, reaching **M1**@22 : 45 and finally arriving **C1**@00 : 45 (+1 day).
- C1** → **M1** → **S1**(08:15-C1S1.08:00): Departing **C1** @ 08:15, reaching **M1**@12 : 00 and finally arriving **S1**@08 : 00 (+1 day).
- C1** → **M1** → **S1**(02:00-C1S1.04:00): Departing **C1** @ 02:00, reaching **M1**@06 : 00 and finally arriving **S1**@04 : 00 (+1 day).
- C1** → **M1** → **S2**(15:00-C1S2.09:45): Departing **C1** @ 15:00, reaching **M1**@18 : 00 and finally arriving **S2**@09 : 45 (+1 day).
- C1** → **M1** → **S2**(09 : 00-C1S2.02 : 15): Departing **C1** @ 09:00, reaching **M1**@11 : 45 and finally arriving **S2**@02 : 15 (+1 day).

Assuming both passengers board a common direct train from the origin point **C1** to either **S1** or **S2** and one of the passengers plans to deboard the train at **M1**. An interesting takeaway from such an itinerary is about zero layover time. But it can have some significant waiting time for passenger who prefer to deboard

out of train. Considering the maximum waiting time at **M1** for deboarding passenger is $\{1, 2, 5\}$ hours. What is the feasible itinerary when:

1. passenger going to **S1** plans to hop out from direct train from **C1** $\rightarrow$ **S2**.
2. passenger going to **S2** plans to hop out from direct train **C1** $\rightarrow$ **S1**.

Scenario E: What is the best possible plan

Now assume that we relax the condition that they both take a common train from **C1**. In this situation, the waiting time for a person starts as soon as the other person leaves **C1**. Also, if a person takes a connecting train (for whatever reason) then he will have a waiting time at the intermediate station too. Then:

1. WAP to print the top 5 travel plans (best travel time is the one with minimum total travel time of both passengers).
2. Add an extra condition that the maximum waiting time for any person (at **C1** + at **M1**) must be less than 10 hours. Print top 5 travel plans.
3. (This is a modification to part 1 and not of part 2). Let W_1 and W_2 denote the total waiting time for persons 1 and 2, respectively. add a condition that $W_1 < \max\{5, 2 \times W_2\}$ and $W_2 < \max\{5, 2 \times W_1\}$. Print top 5 travel plans.
4. (This is a modification to part 2). A person taking a connecting train might want to have some buffer time at the connecting station **M1**; hence, add a minimum layover time of 1 hour at the connecting station. Print top 5 travel plans.